

Technical Session on Retrofitting DUDBC Conference Hall, 22nd November 2016

Participants: Practical Action, IFRC, Norwegian Red Cross, Japanese Red Cross, MoUD CL-PIU, BBC Media Action, UNDP, NRA, Steadfast Nepal, JICA (TPIS-ERP), MOLTIA, Concern, Solve Nepal, Risk Management Office, NSET, Build Change, Habitat for Humanity Nepal, Helvetas, UNOPS, DFID, Lumanti, and Plan.

Introduction

The session was opened by Mr. Parikshit Kadariya, the Deputy Chief of MoUD Central Level Programme Implementation Unit (CL-PIU). Mr. Kadariya highlighted that there has been confusion to date on how the second and third tranches will be released, but the Inspection SOP is now approved and available and set out these processes. He informed that a training package on the Inspection SOP for the DL-PIU engineers is being prepared by MoUD CL-PIU with the NRA, NSET, and JICA and is nearly complete. This is a two day training package and whilst it will be primarily for the DL-PIU engineers who will conduct the inspections, POs' engineers are welcome to participate as they must understand the inspection process also.

Mr. Kadariya informed that a set of retrofitting guidelines, prepared with support from UNDP and led by CoRD and MRB Associated, has been approved by MoUD and will be available on the DUDBC website within a week. The guidelines are divided into three sections, one on RCC structures, one on adobe structures, and one on stone masonry structures. MoUD CL-PIU have kindly made these available for reference in advance of them being published to the DUDBC website, and they can be accessed [here](#). The guidelines includes the principles of retrofitting as well as looking more in depth at the related design calculations. The guidelines do not cover all possible retrofitting methods, but are very comprehensive, and remain a dynamic document that can be added to as other retrofitting methods are developed.

Mr. Kadariya informed that the next technical session will be held on Tuesday 29th November in the DUDBC Conference Hall and will focus on bamboo. It was suggested that there is a need to also include timber, or to have a separate session considering timber. It was highlighted that the technical sessions are very important for sharing experiences and having a common platform to discuss technical issues, challenges and solutions.

Informal Discussion on Retrofitting, Akshaya Paudel, MoUD CL-PIU

This presentation was designed to kick off an exchange of ideas, recognising that Partner Organisations (POs) are interested to know what the GoN guidelines and standards are for retrofitting, but also that the GoN is keen to know what approaches POs have trialled for retrofitting. MoUD CL-PIU informed that the GoN guidelines on retrofitting which will be published on the DUDBC website very soon, may be updated in future to include new or modified technical solutions. The presentation started by clarifying the definitions of repair, restore, and retrofit and stressing that we must be cautious not to mislabel repairs or restorations as retrofits.

The retrofitting methodology covered in the guidelines was presented as a flow chart indicating the different steps and checks required before moving in a particular direction. Retrofitting methods included in the guidelines were also presented. For adobe structures gable wall bands can be added to strengthen the gable, and wire mesh can be anchored to the walls on one or both faces. Where wire mesh is used a good connection between the inside and outside is essential for the wire mesh to work and the mesh should be extended below ground level. Horizontal bands should also be focused on as these are critical areas and retrofitting around openings is extremely important. Polypropylene banding is another option that can be used for retrofitting of adobe structures. Bamboo can also be placed on both sides of the wall and then plastered. This can be a very useful retrofitting option as bamboo is a local material that is widely available in many areas.

For masonry buildings, steel reinforced wire meshing / ferro cement plating, masonry grouting, jacketing on one or both sides of the wall, and the addition of vertical reinforcement are some of the retrofitting methods included in the

guidelines. For RCC buildings, adding shear walls, adding buttress walls, adding and exterior bracing, RC jacketing of columns, steel jacketing of columns, and beam jacketing are some of the retrofitting methods included in the guidelines.

The guidelines also provides a basis for selecting the best method of structural analysis – simple linear, linear, or non-linear – to use and include details on retrofitting foundations. Base isolation and dampers are also included in the guidelines but these are not necessarily relevant for rural reconstruction.

Post-Earthquake Retrofitting in Nepal, Liva Shrestha, Build Change

There is huge potential for retrofit Nepal across the 11 most affected districts, in the Kathmandu Valley, and also in the 17 districts also affected by the earthquakes. There are also a large number of houses that have already been rebuilt but are non-compliant and will need retrofitting to reach compliance, houses in damage grades 1 and 2, and preventative retrofitting which is relevant across the whole country. Build Change estimates that across the 11 most affected districts (excluding the valley districts) 100% of damage grade 3, and 30% of damage grade 4 houses may be retrofitted, a total of 191,657 households based on the CBS survey.

Build Change have conducted a value for money review exercise by comparing the cost of construction for SMM 1.1 design from the DUDBC design catalogue with the cost of retrofitting. It was found that the cost of retrofitting is much less than new construction - 1,410 NPR/ft² for new construction versus 292 NPR/ft² for retrofitting – and that the living space provided by retrofitting was much greater than new construction. Retrofitting was found to save just over 4,000 USD in total, with savings on materials, demolition, transportation, labour, and in the shorter duration of the construction work. The savings for one house were then extrapolated for the 191,657 households estimated to be suitable to retrofit and the savings were found to be as follows:

- 11.49 million metric tonnes less stone required
- 1.53 million man weeks less labour required
- 77.14 billion NPRs (≈0.7 billion USD) reduction in reconstruction costs

However, one of the major challenges to large scale implementation of retrofitting is that it is impossible for structural engineers to design the retrofit for each and every building. To counter this, Build Change are supporting DUDBC with the development of 'type designs' for retrofitting, which would then make the scale up of retrofitting much easier. The main aim is to have a 'type design' approach which would allow any engineer to be able to develop a retrofitting design and submit with the required technical information and implementation package for DUDBC approval. Buildings are categorised into type designs based on factors such as layout, usage, size, materials, etc. This allows for design packages to be developed that can cater to large numbers of buildings that are very similar. This will be supported by go / no go criteria which can be used to decide if a building can be retrofitted, and training for engineers and masons on the retrofitting type designs. Then the inspection process and compliance checklists for retrofitting will also need to be developed.

One example type design which has been developed for typical rural stone and mud mortar masonry buildings was presented. Build Change's design philosophy is to use local, and locally available, materials and methods, make the minimum number of changes to the existing building, and to mobilise the existing strength of the structure to the greatest extent possible.

The typical deficiencies and failures in these rural stone and mud mortar masonry buildings were presented, but also the positive attributes of these traditional structures also. It is these positive attributes that the retrofitting is focused on enhancing. Two variations of this type design for retrofitting have been prepared, one using RCC elements and one using timber elements. Cross / shear walls are not added as the aim of the retrofit is to maintain the flow and the normal usage of the building. The walls are also plastered inside and out to improve the performance of the walls in plane.

The retrofitting process was described as follows:

1. Install ring beam at top of the walls to create positive connection between rafters and the walls. 26 gauge CGI is folded into layers and used as strapping to create this connection.
2. Lateral bracing is added by binding the rafters and purlins with GI wires.
3. Replace the heavy stone gable with a lightweight timber gable.
4. Through ties installed to address delamination.
5. Install strong backs, column type structures (which don't behave as columns) that hold the walls together and prevents walls from falling inside or outside. Installed out of plane at each corner and each pier to take the forces from the wall and prevent out of plane collapse. The corner strong backs grab the walls on both sides.
6. Strengthen floor diaphragm by installing concrete diaphragm at floor level and slab straps to have a positive connection in both directions which facilitates proper diaphragm action.

Build Change proposed that the most critical next steps to scale up retrofitting are to further raise awareness with households, and to develop more type designs. Currently lots of households confuse retrofitting with repair and are not sure what it is, what it means, and how it might be applicable to their home. If more type designs are developed then more buildings can be catered to so that retrofitting can reach scale.

[Experience on Seismic Retrofitting of Buildings in Nepal, Hima Shrestha, NSET](#)

NSET first started retrofitting in Nepal in 1998, and the first building retrofitted was a brick and mud mortar masonry school using a bandage method. Since then, NSET have retrofitted over 300 buildings in Nepal up to 2015, before the Gorkha earthquake. Retrofitting first started in Nepal because there is a need to strengthen vulnerable buildings on a mass scale.

Various methods of analysis were presented and how to compare the existing capacity of a building versus the demand in order to understand what retrofitting requirements are appropriate. The structures capacity after retrofitting can then be analysed to demonstrate the improvement in the structure. For masonry structures non-linear analysis and verification is commonly used for research purposes, as it is time consuming so not practical for a large number of individual structures. In this analysis laboratory testing is used to verify numerical modelling with comparison for different shake levels. Some years back, NSET used bulldozers to carry out model verification as there was no shake table available. The maximum force that the bulldozer could apply found that the retrofitted house was undamaged at double the force at which the unretrofitted house collapsed, but the numerical modelling had shown that the retrofitted structure could sustain forces three times stronger than the unretrofitted house. For RC buildings, pushover analysis is included in NBC and the retrofitting guidelines, and many other countries also use this method of analysis for RC buildings.

When designing retrofitting solutions the performance objective needs to be set first. For some structures such as hospitals, schools, etc. the objective may be immediate occupancy but for all residential buildings the objective will be life safety. If the cost of the retrofit is approximately 30% of new construction costs then it would be considered feasible, however this is a baseline and there is no hard and fast rule. If a retrofit is estimated to cost more than 30% of new construction costs it may still be found to be the best option.

For masonry structures, splint and bandage can be used to provide seismic elements on the outside of the walls where they were not included at the time construction. Jacketing of walls can be done with steel bars or with GI wire mesh. Steel bars require at least 40-50 mm cover so that they don't rust, GI wire mesh requires less generally around 30 mm cover. Depending on the weakness of the building and the design requirements jacketing of walls can be on one or both faces of the wall.

NSET have used polypropylene (PP) band method for adobe buildings prior to the Gorkha earthquake and these buildings performed very well in the earthquake. NSET have also conducted awareness raising on retrofitting using small models on small shake tables.

For areas where cement and steel are not available there are retrofitting options available that use timber sections on either side of the wall and to connect with the wall. One level of testing has already been carried out this in Kunming, and the only problem was local level failure of wall masonry which can be prevented by using gabion wire wrapped around the walls. If timber is not available either, can use gabion wires as a wrap around the walls but also to create bands, this may be applicable for both new construction and retrofit. Gabion wire is not that expensive and can easily be transported to rural areas. Geo mesh may also be an option instead of gabion wire.

For RC buildings, retrofitting options include RC jacketing of columns, where 100mm cover is required and must be carried out for all columns in a structure so more useful for commercial buildings, beam strengthening, which can become very big and expensive, steel profile jacketing, which does not increase stiffness but simply increases strength, and adding a wing wall, which is a very commonly used practice for RC residential buildings. Strengthening of the infill wall is also possible which effectively converts an RC frame to a confined masonry system.

NSET have found that where appropriate retrofitting solutions are developed and implemented the technology can be kept very simple and can be designed by local professionals and can be easily implemented by local technicians after they have received some training. The economic benefits of retrofitting are extremely important with retrofitting normally 20-30% of reconstruction costs. Retrofitting also does not require hard to access materials, and can be fully based on locally available materials. However, they have also found that to date the replication rate is very slow for retrofitting and often this is due to the disruption in the use of building whilst the retrofitting is being carried out and a lack of awareness on the options available. There is a need to do continuous research and implementation to keep improving and finding ways to bring retrofitting to scale.

Retrofitting in Nepal – the Case and Rationale, Magnus Wolfe-Murray, DFID

This presentation looked at retrofitting from the perspective of social and heritage preservation. Examples were provided from Sindhuli where there are many houses that fall into damage category three and households are unsure about how to proceed as it will cost them an awful lot of money to demolish and rebuild, but they do not have sufficient information on retrofitting to make an informed decision on this. There are also social and heritage considerations, the traditional rural homes use local materials and building practices and include space for storage at the top, living in the middle, and animals on the ground floor. If these homes are demolished and rebuilt using different materials and designs there will be a huge social and cultural loss, and there will be the economic factors of the high costs involved in demolishing and rebuilding.

The comparative cost analysis exercise presented by Build Change demonstrated just how big the implications of this are, given that the total savings across the reconstruction could be close to 2 billion USD. A damage level map prepared by the HRRP was presented and it was explained that in the districts where there are very high levels of complete collapse there is training and technical assistance being provided to support new construction but in the more Southern part of the map the vast majority of houses were not fully damaged and there is scope for TA and training to be more focused on retrofitting. In areas where new construction is the focus there is also a need to consider that up to 200,000 houses have already been rebuilt (this is based on anecdotal findings across Northern Gorkha, Dhading, Dolakha, etc.) and will need support to retrofit to meet compliance criteria. Photos from Tippling in Northern Dhading were presented which were taken in May 2016 where more than 80% of houses had already rebuilt their homes at that point with little or no application of safe building practices. A similar situation was presented from Haku in Rasuwa district.

Photos were also shown from Northern Dhading with examples of excellent construction work, with great corner stones, nice stone masonry, and nice slate. But if these houses are part of the caseload of category 5, there will be a need to look at retrofitting these so they are compliant. The households in these areas had said that they will not take it down to start again with improvements but the earthquake showed that these types of buildings all collapsed and will need to be strengthened to be resilient.

It was also highlighted that there is a need to consider the other 17 districts also. DFID are currently supporting the NRA to carry out the damage survey in these districts, with the survey starting in the next month and running for 1-2 months. The NRA currently estimates that there are 108,000 damaged houses across these districts but it is not sure what category of damage. It can be expected that many of these houses are probably rebuilt already and will likely have compliance issues so there is a need to look at how these houses can be supported to be compliant. It would be good to clarify what resources are available for the 17 districts. There is clearly a need to support across these districts also and we must look at how to effectively support in these areas.

Questions and Discussion

Q: For the PP banding, how do the connections work? With steel jacketing, the pins connect through the wall quite solidly. With PP how are they connected and how is it held on the wall?

- The PP is used on both sides of wall and drill through the wall at certain spacing and like stitching, the two horizontal and vertical bands are stuck together. At certain spacing there is through connection and then at other spacing up to a certain depth in the wall only.

Q: When geo fabrics are used for reinforcing and bands is there a bonding problem with mortar and plastic? Is it slippy in comparison to steel or wire?

- NSET have not been using this method so far but where it has been used for adobe buildings it performs well and this connection is not a problem. UV may be problem but so long as it is plastered then it will be ok. Even for PP band can use mud plaster.

Q: If any PO intends to do retrofitting training are there guidelines / curriculum to follow?

- MoUD CL-PIU have a syllabus for training, and POs interested in delivering in training can follow the [NRA's training management and facilitation guidelines](#) to apply for and to manage training. MoUD CL-PIU would recommend that it is better to conduct trainings for technicians initially. The retrofitting training package is already developed, with the support of the Institute of Engineers, and is available to all interested POs. The package is four days, but could be extended to five days as it is very intense.

Q: Are you aware of any trainings that are being conducted on retrofitting by organisations? Or are there any organisations that are preparing to conduct them?

- MoUD CL-PIU informed that to date, no organisation has permission to conduct trainings on retrofitting and no organisations have applied to deliver such training. DUDBC has conducted a three hour session on retrofitting for DUDBC technicians but the full training package is 4-5 days and all the technicians and engineers will be required to complete this training soon, but first priority will be given to the inspection SOP training, and the retrofitting training will take place after.

Q: If a PO was intending to do some training and wanted approval from DUDBC for a different type of curriculum how would this work?

- The [NRA's training management and facilitation guidelines](#) provide for two ways to get approval; one with existing curriculum and another where a new curriculum is proposed. Where a new curriculum is proposed there is a committee in MoUD CL-PIU that will review it and if it found to be according to requirements the committee will give permission for this training to go ahead. Any new curriculum need to be submitted with a thorough explanation of how the curriculum has been developed. MoUD CL-PIU have already provided permission for slightly different masons training, so the provision is there.

Q: If a POs existing project does not include retrofitting, but they now wish to include retrofitting training, based on the approved guideline and using curriculum, can they do this or is a separate project agreement required?

- MoUD CL-PIU said this was a difficult question to answer, and it's something that would need to be discussed with the NRA also, but it was felt that where changes / additions are for a good purpose it would be possible to find an arrangement to enable this.

Q: Are there any initiatives to encourage people to retrofit rather than reconstruct their homes?

- MoUD CL-PIU answered on behalf of the NRA and informed that the NRA is preparing grant disbursement guidelines for maintenance / rehabilitation, where a certain amount, around 50,000 NPRs, will be provided. Retrofit is a very technical word, requires engineering judgement and design. The guidelines for this are currently in the cabinet for endorsement.

Q: Is 100,000 NPRs considered sufficient for repair?

- The housing grant for new construction, of 300,000 NPRs, is not provided to fully cover the cost of new construction and the approach for repair is the same. The intention is to incentivise compliance.

Q: Will a communication strategy be developed for retrofitting to support consistent messages and information are shared with the affected communities?

- MoUD CL-PIU suggested that it may be useful to have a workshop on this, and that the HRRP could support with this. There is certainly a need to prepare more IEC materials and to have more practical demonstration of retrofitting so that people can see how it works.
- UNDP informed that a series of ten posters based on the guidelines is being developed and is expected to be completed and shared soon.

Q: Would it be possible to prepare a series of FAQs, from a non-technical perspective, that could be shared with households?

- The HRRP is currently preparing a first version of a 'Q&A' document with a mix of questions covering many aspects of the reconstruction. The document will be a living document that will be updated regularly and will rely on partners to submit questions, and ideally it should become more focused on questions from households and communities.

Build Change informed that one of the most striking things from their experience is that the need is so great at so many levels, but then in the affected areas people are not aware of retrofitting. This is probably the most important building season since the earthquake and a lot of ppl are going to make key decisions about what to do. One option for POs could be to review their agreement with the NRA so that where demonstration house construction is included, and there are many houses in their working areas that are partially damaged, they could look to include demonstration retrofits instead. This would be a very useful way for engineers / masons to learn how to do it, and a way for communities to be aware of the options available to them. Build Change have also found that in a VDC there is a tendency for people to think that retrofitting is a solution for poorer households and everyone seems to want an RC frame and brick house. In order to overcome this, POs should try to find influential, wealthy people in the community for the demonstration retrofits as it sets an example that retrofitting is a positive thing, and will encourage others to adopt the approach. MoUD CL-PIU clarified that the reconstruction committees will be responsible for selecting the household to receive demonstration construction / retrofit and it will be important to look at how to address this to make the most of demonstration activities. MoUD CL-PIU are very enthusiastic about the concept of demonstration retrofits and requested that interested partners submit proposals and they will work to provide approvals as soon as possible.

DFID highlighted that the MoUD DL-PIUs have huge numbers of engineers, sub-engineers, and technicians in the districts and at VDC level, and POs should team up with these engineers to really maximise the potential for retrofitting.

MoUD CL-PIU informed that an MIS is being developed which includes a component on monitoring of POs activities and this will be presented soon. In the interim, POs are requested to report regularly to MoUD CL-PIU and reports can be sent via email to moudclpiu@gmail.com.

The HRRP closed the meeting with the reminder that the vast majority of POs are not involved in the housing grant disbursement and are focusing on technical assistance to facilitate and support the GoN reconstruction programme.

It is up to everyone to work together to bring retrofitting to the fore as the money that will be saved is the GoN and affected households' money and there is a responsibility on all actors in the reconstruction to support with this.

Next Steps

- POs with demonstration houses included in their programmes, who are interested at looking to replace these with retrofit demos in appropriate areas are requested to contact Loren, the HRRP National Coordinator (loren.lockwood@hrrpnepal.org), with information on the number and location of potential retrofit demos so that a collective strategy can be developed and presented to the NRA and MoUD CL-PIU. The HRRP may also be able to support with linking interested partners with training opportunities and exposure visits for their technical staff.
- HRRP to support with development of communications strategy on retrofitting by conducting workshop with GoN, POs, and relevant technical experts to determine key technical and policy information messages to be shared to households and communities.

Reference Documents

- GoN retrofitting guidelines for RCC, adobe structures, and masonry structures are available for information [here](#). These will be published on the DUDBC website soon.
- UN Habitat Experience in Repair and Retrofitting in the Housing Sector after the Kashmir Earthquake, Pakistan 2005, available [here](#).
- Nepal National Building Code available on the DUDBC website [here](#).

Contact Details

Contact details for the teams presenting at the technical session are provided below in order to facilitate follow up on any of the topics covered in the session.

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